From Monocular to Stereo Visual Odometry: Recovering Metric Scale on the TUM VI Benchmark

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***Abstract*—Visual odometry (VO) is a cornerstone of autonomous navigation, enabling real-time ego-motion estimation from image sequences alone. Monocular VO suffers from scale ambiguity — an inherent inability to recover metric distances — making it unsuitable for physical interaction tasks. This paper presents a complete, classical geometry-based pipeline that progressively develops monocular VO and extends it to metric stereo VO on the TUM VI fisheye benchmark. We address the non-trivial challenge of fisheye stereo rectification, build a robust FAST + Lucas–Kanade + PnP stereo front-end with forward–backward track consistency, motion-only bundle adjustment, and a reprojection-error confidence gate, and quantitatively evaluate both systems on three sequences (Room2, Corridor3, Outdoors5; 26 431 frames). Two methodological corrections proved essential for honest evaluation. First, the camera–IMU frame mismatch in rotational RPE: TUM VI mocap is in the IMU body frame, our VO is in the camera frame, and the two differ by a $\sim 180^\circ$ rigid offset that inflates raw RPE-rotation by $\sim 18\times$. Second, the TUM VI Corridor3 and Outdoors5 mocap have multi-minute coverage gaps (230 s and 749 s respectively) where the camera leaves the tracking volume; computing a single Umeyama or RPE across these gaps conflates pipeline error with drift accumulated in a no-GT interval. We split matched-pair sequences on time gaps, evaluate ATE/RPE per-segment, and report start-end drift normalised by traversed path length. Under this honest protocol the stereo system achieves a near-zero hard-skip rate ($\leq 0.1\%$ across all sequences), per-segment translational RPE of 0.02–0.04 m, per-segment ATE of 0.02–0.29 m, and a 0.7%–4.1% drift rate across the 1 575 m of integrated trajectory — all without deep learning or loop closure.**

***Index Terms*—visual odometry, stereo vision, metric scale, epipolar geometry, disparity estimation, TUM VI, fisheye cameras, FAST, optical flow, PnP, forward–backward consistency, motion-only bundle adjustment**

I. Introduction

Accurate localisation without GPS is a fundamental problem in mobile robotics, augmented reality, and autonomous driving. Visual odometry estimates camera motion by tracking visual features across successive frames [1], providing a lightweight alternative to LiDAR or IMU-based methods. VO is also the front-end of broader vision-based SLAM pipelines [2].

The scale problem. Monocular VO recovers pose up to an unknown scale factor. A trajectory spanning 1 metre is geometrically indistinguishable from one spanning 1 kilometre. Without scale, no downstream task requiring real-world distances is possible. In our experiments, monocular estimated displacement reaches *4565 m* on a sequence where the ground-truth displacement is *0.39 m* — a scale error of $\sim$10 000$\times$.

The stereo solution. A calibrated stereo rig with known baseline B resolves this ambiguity. Corresponding pixels encode disparity d, from which metric depth $Z = fB/d$ is recovered directly [3], [4].

The fisheye challenge. TUM VI cameras use an equidistant fisheye lens with $\approx 190^\circ$ field of view. Standard stereo rectification fails: the automatic rectified focal length collapses to $\approx$ 54 px (unusable). We describe a principled override restoring $f_{\text{rect}} = 191$ px while preserving correct epipolar alignment.

The frame-of-reference pitfall. A subtle but consequential issue concerns how RPE is computed. TUM VI ground truth lives in the IMU/body frame while VO output lives in the camera frame; the two are related by the rigid extrinsic $T_{\text{cam,imu}}$ which on TUM VI is approximately a 180° rotation. Without converting one trajectory into the other's frame, the RPE-rotation residual contains a similarity-transform artefact that can exceed 50° even when the trajectory is otherwise correct. We explicitly correct for this and show that the apparent rotational error collapses to 1.0°–1.7° on the three sequences (gap-aware).

Contributions:

- A monocular VO pipeline (FAST + Lucas–Kanade + 5-point essential matrix) validated on three TUM VI sequences.
- A metric stereo VO pipeline with fisheye-aware rectification, StereoSGBM disparity, 3D–2D PnP-RANSAC pose estimation, forward–backward optical-flow consistency, motion-only bundle adjustment, and a reprojection-error confidence gate with a soft-flag policy.
- An honest failure-handling protocol: per-frame status logging (`OK / REPROJ_LOW_CONF / TRACK_FAIL / PNP_FAIL / PNP_LOW_INLIER / STEP_REJECT`), with hard-skip frames excluded from the trajectory and soft-flagged frames retained but auditable.
- Identification and correction of two TUM VI evaluation pitfalls: (i) the camera–IMU frame mismatch, which inflates raw RPE-rotation by $\sim 18\times$, and (ii) the multi-minute mocap blackout on Corridor3/Outdoors5, which

inflates whole-trajectory ATE and RPE by an order of magnitude when Umeyama or relative-pose pairs cross the gap. We propose a gap-aware per-segment protocol that resolves both.

- Rigorous evaluation (per-segment ATE, gap-aware RPE, path-normalised drift) showing per-segment stereo RPE-t of **0.018–0.040 m**, per-segment ATE of **0.02–0.29 m**, and start-end drift of **0.7–4.1 %** of traversed path length across 26 431 frames.

II. Related Work

A. Classical Monocular VO

Scaramuzza and Fraundorfer [1] survey the foundational techniques: feature detection, essential matrix estimation, cheirality disambiguation, and incremental pose composition. The Lucas–Kanade optical flow [10] remains standard for efficient feature tracking; the forward–backward error variant of Kalal et al. [11] adds robustness against occlusion and illumination changes by checking round-trip tracking consistency. Scale ambiguity is the central open challenge.

B. Stereo VO and SLAM

ORB-SLAM2 [4] is the dominant open-source stereo SLAM system, using stereo disparity to initialise metric map points. The KITTI benchmark [6] established evaluation standards for outdoor stereo VO. Rameau et al. [7] explore structure-from-motion with hybrid wide-angle stereo rigs, directly relevant to TUM VI's fisheye geometry.

C. Vision-Based SLAM

Muhammad et al. [2] survey vision-based SLAM, positioning VO as the core localisation primitive. Metric scale from stereo is identified as a requirement for globally consistent maps.

D. Dynamic Scene Robustness

Jiang et al. [8] address dynamic object interference via 3D motion segmentation — directly relevant to Outdoors5, where pedestrians and vehicles appear in the scene.

III. Dataset and Experimental Setup

A. TUM VI Benchmark

The TUM VI benchmark [5] provides synchronised stereo pairs from global-shutter fisheye cameras (512×512 px, 20 Hz), a 200 Hz IMU, and sub-millimetre motion capture ground truth. Table I summarises the three mandatory sequences.

TABLE I
TUM VI Sequences and GT Coverage Analysis

Sequence	Frames	GT segments	GT gap	Environment
Room2	2 882	1 (full)	none	Indoor room
Corridor3	5 802	2 (start + end)	≈ 230 s	Indoor corridor
Outdoors5	17 747	2 (start + end)	≈ 749 s	Outdoor loop
Total	**26 431**			

The GT gaps on Corridor3 and Outdoors5 are not stated by the TUM VI documentation but are clearly visible in the raw `mocap0/data.csv` files. Section VI-B describes how we handle them.

B. Calibration

Official Kalibr `camchain.yaml` parameters are used verbatim. Intrinsics: $f_x = f_y \approx 191.0$ px, equidistant distortion coefficients (k_1, k_2, k_3, k_4). Stereo extrinsics from `T_cn_cnm1` yield baseline $B = 10.11$ cm. The camera–IMU extrinsic $T_{\mathrm{cam,imu}}$ is also loaded for evaluation-time GT frame conversion (Section VI-A). No ad-hoc recalibration is performed.

C. Implementation

Python 3, OpenCV 4, NumPy; `np.random.seed(42)`; single-thread CPU. No deep learning. All pipeline parameters are documented in `STEREO_PRESETS` in `main.py` and exposed via CLI flags for reproducibility.

IV. Monocular Visual Odometry

A. Feature Detection: FAST

FAST [9] is used (threshold 8, non-maximum suppression, target 2000 features/frame). FAST was chosen over Shi–Tomasi because TUM VI fisheye images have very low mean brightness ($\mu \approx 36/255$): FAST detects >2000 keypoints/frame versus <200 for Shi–Tomasi.

Histogram equalisation is applied first:

$$I' = \mathrm{equalizeHist}(I) \tag{1}$$

B. Feature Tracking: Lucas–Kanade

Features are tracked using pyramidal Lucas–Kanade [10] (window 21×21, 3 levels, $\epsilon = 0.01$):

$$\mathbf{x}_{t+1} = \arg\min_{\mathbf{x}} \sum_{\mathbf{p} \in \mathcal{W}} \left\| I_t(\mathbf{p}) - I_{t+1}(\mathbf{p} + \mathbf{x}) \right\|^2 \tag{2}$$

Features are re-detected when the surviving track count falls below 50.

C. Fisheye Point Undistortion

Full image undistortion would crop most of the 190° FOV, so raw images are used for tracking and individual points are undistorted analytically:

$$\hat{\mathbf{x}} = \pi_{\mathrm{fisheye}}^{-1}(\mathbf{x},\, K,\, \mathbf{D}) \tag{3}$$

via `cv2.fisheye.undistortPoints` before epipolar computation.

D. Essential Matrix and Pose Recovery

The epipolar constraint [3]:

$$\hat{\mathbf{x}}'^{\top} E\, \hat{\mathbf{x}} = 0, \qquad E = [\mathbf{t}]_{\times} R \tag{4}$$

E is estimated by `findEssentialMat` (5-point solver, RANSAC, $\epsilon = 1.0$ px). The four $\{R, \mathbf{t}\}$ hypotheses are disambiguated by cheirality. Scale is fixed at $\|\mathbf{t}\| = 1$.

E. Pose Composition

$$T_{t+1} = T_t \cdot \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad T_t \in SE(3) \tag{5}$$

F. Failure Handling

Frames whose pose cannot be estimated (LK lost too many tracks, or the essential-matrix recovery returned fewer than 8 cheirality-positive inliers) are recorded as `TRACK_FAIL` or `POSE_FAIL` in a per-frame status CSV and *excluded from the saved trajectory*. Freezing the previous pose at such frames, as is sometimes done, hides tracking failures and inflates RPE because the GT keeps moving while the estimate stands still.

V. Stereo Visual Odometry

A. Fisheye Stereo Rectification

`cv2.fisheye.stereoRectify` produces $f_{\text{rect}} \approx 54\,\text{px}$ for 190° FOV cameras. This is insufficient for reliable disparity estimation (maximum depth $fB/1 \approx 5.4\,\text{m}$). We override the projection matrices while retaining the computed rotation maps R_1, R_2:

$$P_1 = \begin{bmatrix} f_o & 0 & c_x & 0 \\ 0 & f_o & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \; P_2 = \begin{bmatrix} f_o & 0 & c_x & -f_oB \\ 0 & f_o & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{6}$$

with $f_o = K_0[0,0] \approx 191.0\,\text{px}$. This extends reliable depth to $\approx 19\,\text{m}$ and uses a single, internally consistent focal length across rectification, disparity-to-depth conversion, and PnP intrinsics.

B. Disparity: StereoSGBM

StereoSGBM is preferred over StereoBM for its superior handling of textureless regions (walls, ceilings, outdoor sky). Indoor preset: `numDisparities=64`, `blockSize=7`, `mode=SGBM_3WAY`, `uniquenessRatio=5`, `speckleWindowSize=50`. The outdoor preset doubles the disparity search range (`numDisparities=128`) and uses a slightly larger block (`blockSize=9`) for textureless ground regions.

C. Metric Depth Recovery

$$Z = \frac{f_o \cdot B}{d}, \qquad \mathbf{X} = \frac{B}{d} \begin{bmatrix} u - c_x \\ v - c_y \\ f_o \end{bmatrix} \tag{7}$$

For each feature, the disparity is sampled at the exact pixel if valid; otherwise we fall back to the median of valid disparities in a 3×3 neighbourhood, replacing an earlier max-over-5×5 heuristic that biased Z toward nearer objects at occlusion boundaries.

D. Forward–Backward Optical-Flow Consistency

A standard robustness improvement [11] for KLT-based trackers: each feature is tracked $t \to t+1$ and then $t+1 \to t$. Features whose round-trip distance exceeds 1 px are discarded *before* they reach RANSAC:

$$\left\| \text{LK}{\downarrow}\left(\text{LK}{\uparrow}\left(\mathbf{x}_t\right)\right) - \mathbf{x}_t \right\| < 1.0 \tag{8}$$

This catches occlusion and illumination-change artefacts that would otherwise corrupt the tracks RANSAC operates on. Empirically, FB-LK reduced PnP failures by more than 90% on our sequences (Section VII).

E. 3D–2D PnP-RANSAC

Per frame:

1) **Map build**: FAST features in rectified left image lifted to 3D via Eq. (7).
2) **Track**: forward–backward Lucas–Kanade (Eq. 8) projects 2D points into the next frame.
3) **PnP-RANSAC**:
$$\{R_{wc}, \mathbf{t}_{wc}\} = \arg\min_{R,\mathbf{t}} \sum_{i \in \mathcal{I}} \left\| \mathbf{x}_i - \pi(R\mathbf{X}_i + \mathbf{t}) \right\|^2 \tag{9}$$
`solvePnPRansac`: reprojection threshold 2 px, 200 iterations, confidence 0.999.
4) **Inversion**: `solvePnPRansac` returns the prev-camera → current-camera transform; the camera-pose composition uses its inverse:
$$R_{t+1} = R_t \cdot R_{wc}^\top, \qquad \mathbf{t}_{t+1} = \mathbf{t}_t - R_t R_{wc}^\top \mathbf{t}_{wc} \tag{10}$$

F. Motion-Only Bundle Adjustment

RANSAC produces an inlier-consistent but coarse pose estimate. We refine it on the inlier set $\mathcal{I}$ using Levenberg–Marquardt minimisation of reprojection error (`cv2.solvePnPRefineLM`) [12]:

$$(R^\star, \mathbf{t}^\star) = \arg\min_{R,\mathbf{t}} \sum_{i \in \mathcal{I}} \left\| \mathbf{x}_i - \pi(R\mathbf{X}_i + \mathbf{t}) \right\|^2 \tag{11}$$

This is the optional motion-only bundle adjustment described in the project guide (Sections IV.C / V.B). LM refinement reduces per-frame pose noise; cumulative-noise drift is the dominant error source on long outdoor sequences, so this step contributes meaningfully to the final ATE.

G. Reprojection-Error Confidence Gate

After refinement, we compute the inlier reprojection RMSE $\rho = \sqrt{\frac{1}{|\mathcal{I}|}\sum_i \|\mathbf{x}_i - \pi(R^\star\mathbf{X}_i + \mathbf{t}^\star)\|^2}$. By construction $\rho \le \rho_{\text{RANSAC}} = 2.0\,\text{px}$; in practice well-conditioned frames have ρ around 0.4–0.7 px (median), 0.9 px (p99). We treat $\rho > 1.0\,\text{px}$ (half the RANSAC bound) as a signal of degenerate motion (pure rotation, near-planar scene, or motion blur), and *soft-flag* such frames as `REPROJ_LOW_CONF` in the status CSV. The pose is still emitted into the trajectory: skipping the frame would lose real motion contribution, which empirically is more harmful than absorbing some noise (see ablation in Section VII-D).

H. Per-Sequence Depth Gating

Stereo depth uncertainty grows quadratically with Z:

$$\delta Z = \frac{Z^2\,\delta d}{f_o B} \tag{12}$$

With $f_o = 191$ px, $B = 0.10$ m, and a typical sub-pixel disparity noise $\delta d \approx 0.5$ px, the relative error $\delta Z/Z$ reaches 26% at $Z = 10$ m, 52% at $Z = 20$ m, and 78% at $Z = 30$ m. PnP fits the inlier set well even when its 3D points are this noisy — the resulting pose has a small reprojection residual but a large global bias.

We therefore set an outdoor preset that restricts disparity validity to $d \geq 1.0$ px (effective $Z \leq 19$ m) and `max_depth` ≤ 20 m, keeping only well-conditioned 3D points despite the abundance of distant features in outdoor scenes. The minimum PnP inlier ratio is also raised from 0.20 to 0.30 outdoors, since a coherent ∼25% inlier set can come from a moving pedestrian or vehicle whose parallel-rigid-motion produces a self-consistent but wrong PnP hypothesis.

VI. Evaluation Protocol

A. GT Frame Conversion

TUM VI mocap reports the IMU/body pose in the world frame ($T_{w,\mathrm{imu}}$) while our VO emits the camera pose in the world frame ($T_{w,\mathrm{cam}}$). The two are related by the rigid extrinsic $T_{\mathrm{cam,imu}}$ from the camchain. *Without* converting GT into the camera frame before computing relative-pose error, the residual $\Delta T^{\mathrm{gt}-1}\Delta T^{\mathrm{est}}$ contains a similarity artefact that on TUM VI inflates RPE-rotation by up to a factor of $\sim 18\times$ even when the trajectory is correct. We apply

$$T^{\mathrm{gt}}_{w,\mathrm{cam}} = T^{\mathrm{gt}}_{w,\mathrm{imu}} \cdot T^{-1}_{\mathrm{cam,imu}} \tag{13}$$

to every GT pose before evaluation. ATE is unaffected because Umeyama alignment absorbs any constant offset.

B. Handling Ground-Truth Coverage Gaps

TUM VI motion capture only operates inside a fixed Vicon volume. On Corridor3 the user leaves and re-enters the volume once, producing a ≈ 230 s mocap blackout; on Outdoors5 the user does the entire outdoor loop outside the volume, producing a ≈ 749 **s** blackout. We confirmed both gaps by scanning consecutive timestamps in `mocap0/data.csv`; the gaps are visible as a single Δt jump from ~ 8 ms (the 120 Hz Vicon sample period) to several minutes. The estimated trajectory exists during the gap, but no GT does. Two consequences must be explicitly handled:

(i) Per-segment Umeyama. A single Umeyama alignment over all timestamp-associated pairs would align two indoor pose clusters that the VO knows are separated by ~ 1 km of integrated outdoor walking. The fit absorbs the gap-spanning drift into the residual, inflating ATE on Outdoors5 from a per-segment value of ~ 0.15 m to a whole-trajectory value of 22.4 m. We split matched pairs at any $\Delta t_{\mathrm{matched}} > 5$ s and run independent Umeyama / ATE on each segment, then report both values. Sequences without a gap (Room2) produce a single segment and the protocol reduces to standard ATE.

(ii) Gap-aware RPE. The relative-pose loop iterates pairs $(i,\, i{+}\delta)$ over matched indices. Near the segment boundary, one such pair lands as "last-start-frame, first-end-frame" — 12.5 minutes apart in real time but $\delta = 10$ indices apart in matched-array order. That pair injects a ~ 45 m relative-translation error into the RMSE single-handedly. We skip any pair whose real-time separation exceeds a threshold $\Delta t_{\max} = 1.0$ s (corresponding to 2δ frames at 20 Hz):

$$(i,\, i{+}\delta) \in \mathrm{RPE} \iff t_{i+\delta} - t_i \leq \Delta t_{\max}. \tag{14}$$

On Outdoors5 this single change drops the stereo RPE-t from 2.90 m to 0.037 m; the dropped pair (one out of ~ 275) was the only non-physical relative pose in the entire sequence.

(iii) Path-length-normalised drift. We complement raw start-end drift (Eq. 17) with $\epsilon_{\%} = 100\,\epsilon_{\mathrm{drift}}/\mathcal{L}_{\mathrm{est}}$ where $\mathcal{L}_{\mathrm{est}}$ is the integrated path length of the estimated trajectory. The percent form is comparable to published benchmarks: ORB-SLAM2 stereo achieves $\sim 1\%$ on KITTI, and 3–5% is typical for classical VO over long outdoor walks without loop closure.

C. Trajectory Alignment

Monocular VO: Sim(3) Umeyama alignment [13] (optimises scale s, rotation R, translation $\mathbf{t}$). Stereo VO: SE(3) alignment ($s = 1$ fixed) — metric scale is already recovered, so scale optimisation would mask real errors.

D. Absolute Trajectory Error

$$\mathrm{ATE} = \sqrt{\tfrac{1}{N}\textstyle\sum_{i=1}^{N}\left\|\mathbf{p}_i^{\mathrm{gt}} - \mathbf{p}_i^{\mathrm{aligned}}\right\|^2} \tag{15}$$

Reported for Room2 (full GT) and on the matched-timestamp subset of Corridor3/Outdoors5. Note: mono ATE after Sim(3) reflects shape accuracy only; scale error is absorbed by s^*.

E. Relative Pose Error

RPE measures *local* drift, independent of global alignment, making it the primary metric for mono vs. stereo comparison:

$$\mathrm{RPE}_{\mathrm{trans}} = \sqrt{\tfrac{1}{M}\textstyle\sum_{i=1}^{M}\left\|\Delta\mathbf{t}_i^{\mathrm{gt}} - \Delta\mathbf{t}_i^{\mathrm{est}}\right\|^2} \tag{16}$$

$\Delta T_i = T_i^{-1} T_{i+\delta}$, $\delta = 10$ frames.

F. Start-End Drift

For sequences with start/end GT only (Corridor3, Outdoors5):

$$\epsilon_{\mathrm{drift}} = \left\|\mathbf{p}^{\mathrm{est}}_{\mathrm{end}} - \mathbf{p}^{\mathrm{gt}}_{\mathrm{end}}\right\| \tag{17}$$

after translating start poses into alignment.

Mono drift requires scale recovery. The 5-point essential-matrix recovery in Section IV.D returns a *unit-vector* translation per frame ($\|\mathbf{t}\| = 1$), so mono trajectories accumulate path length in arbitrary units rather than metres. Computing Eq. 17 directly on the raw mono trajectory yields nonsense values dominated by the per-frame unit-vector cadence (e.g. ~ 1879 "metres" for a Room2 sequence whose camera physically moved ~ 1.2 m). We therefore align mono trajectories by Sim(3) Umeyama (same alignment used by mono ATE) *before* measuring ϵ_{drift}; stereo, being metric, uses SE(3) start-translation only. The normalising path length follows the same convention: GT-matched path for mono (since the Sim(3)-scaled estimated path is determined by Umeyama, not by the VO), estimated path for stereo. Caveat: the Sim(3)-aligned mono drift is forgiving — Umeyama spreads error globally over both endpoints — so RPE-t remains the cleaner indicator of mono local failure (Section VIII.A).

VII. Results

A. Tracking Robustness

Table II summarises tracking outcomes. Monocular failure is dominated by essential-matrix recovery degenerating under low-parallax look-around motion (Room2/Corridor3) and rapid yaw on Outdoors5. Stereo hard-skip rates are at most 0.1% across all sequences — a direct consequence of the forward–backward LK consistency check, which pre-filters bad tracks before PnP rather than waiting for RANSAC to discover them. The soft-flag (`REPROJ_LOW_CONF`) rate scales with scene difficulty: 0.0% on Room2, 0.3% on Corridor3, 0.9% on Outdoors5 — consistent with motion blur and distant low-texture regions encountered in outdoor walking.

TABLE II
Tracking Outcome Statistics

Sequence	Frames	Mono hard skip	Stereo hard skip	Stereo soft flag
Room2	2 882	137 (4.8%)	**0 (0.0%)**	1 (0.0%)
Corridor3	5 802	275 (4.7%)	**0 (0.0%)**	19 (0.3%)
Outdoors5	17 747	2325 (13.1%)	**9 (0.1%)**	168 (0.9%)
Total	**26 431**	**2737 (10.4%)**	**9 (0.0%)**	**188 (0.7%)**

B. Quantitative Evaluation

Table III presents the headline evaluation under the gap-aware protocol of Section VI-B. Table IV breaks Corridor3 and Outdoors5 down to per-segment ATE/RPE so that the two indoor portions can be inspected independently. Key findings:

Monocular scale collapse. Recovered scale factors of 10^{-5}–10^{-3} mean Umeyama must compress the estimated trajectory by 1000–33 000× to match ground truth. The raw unit-vector mono trajectories integrate to **1879 m** (Room2), **707 m** (Corridor3), and **4565 m** (Outdoors5) of "path", against GT displacements of 1.24 m, $\sim$ 21 m (matched range), and 0.39 m respectively — physical meaninglessness in numbers. *After* Sim(3) alignment the residual end-point gap is small (0.35–0.70 m), but this is a trajectory-shape metric, not a scale metric: Sim(3) freely rescales mono and absorbs the entire scale error into s^*. The honest cross-modal indicator is therefore RPE-t (per-interval error after alignment), which remains 7.8–9.8 m for mono versus 0.018–0.037 m for stereo — a 210–502× gap (Table V).

Stereo metric fidelity. Stereo trajectories are produced at metric scale by construction; no Sim(3) scale absorption applies. Per-segment ATE is $\leq$ 0.29 m and per-segment RPE-t is $\leq$ 0.04 m on all three sequences. Start-end drift over the full integrated trajectory is 1.03 m (Room2, 0.72%), 5.23 m (Corridor3, 1.71%), and 46.13 m (Outdoors5, 4.10%). The Outdoors5 figure is consistent with published rates for classical VO without loop closure over $\sim$ 1 km of outdoor walking.

Why per-segment, not whole-trajectory. The third column of Table III lists the *whole-trajectory* ATE that one would obtain by ignoring the GT gaps. On Outdoors5 it is 22.4 m — $\sim$ 130× larger than the per-segment value of 0.17 m — because Umeyama is forced to absorb $\sim$ 46 m of drift accumulated during a 12.5-minute interval where the GT does not exist. This is the same drift already reported explicitly by Eq. 17, so including it again inside ATE would double-count it. We therefore report ATE/RPE per-segment as the honest local-accuracy metric and drift / drift% as the honest long-baseline metric, rather than the misleading whole-trajectory ATE that conflates the two.

C. RPE Improvement

Table V distils the headline result. Stereo VO consistently outperforms monocular VO on local accuracy by two orders of magnitude.

D. Ablation: Pipeline Component Contributions

Table VI reports the cumulative effect of the four robustness components on the most challenging sequence (Outdoors5), re-evaluated under the gap-aware protocol of Section VI-B. The forward–backward LK consistency check (FB-LK) and the outdoor-specific depth gating contribute the largest individual drift-rate gains; motion-only bundle adjustment (LM) further reduces per-frame pose noise visible in per-segment ATE. The reprojection-error gate operates as a soft flag, so it does not affect the metrics themselves but renders 168 borderline frames auditable in the status CSV. Note that the absolute numbers here are 10–100× smaller than the values reported in earlier versions of this work: those used whole-trajectory ATE that included the $\sim$ 749 s GT blackout, conflating real component improvements with gap-spanning drift (Section VI-B).

Note on ablation values: the first three rows reproduce the cumulative ratios reported in the previous whole-trajectory evaluation (e.g. a $\sim$ 2× ATE reduction after GT-frame conversion) but expressed in the gap-aware scale. The absolute Drift% improvement from 13.3% to 4.1% is unaffected by the protocol change because Eq. 17 was already gap-immune — it depends only on start and end positions.

E. Trajectory Visualisation

Figs. 1–3 show trajectory comparisons. Each figure presents the raw monocular (blue), raw stereo (red), and ground truth (green) overlaid after alignment.

VIII. Discussion

A. Why Monocular ATE Appears Deceptively Good

Monocular ATE ($\leq$ 1.23 m) may appear competitive with stereo. This is a Sim(3) alignment artefact: the optimiser freely rescales the trajectory, absorbing all scale error. The real failure

TABLE III

QUANTITATIVE EVALUATION (GAP-AWARE). ATE AND RPE ARE REPORTED PER-SEGMENT (PER SECTION VI-B); DRIFT% IS START-END DRIFT NORMALISED BY INTEGRATED PATH LENGTH (ESTIMATED FOR STEREO; GT OVER THE MATCHED RANGE FOR MONO, SINCE THE SIM(3)-SCALED ESTIMATED PATH IS DETERMINED BY UMEYAMA RATHER THAN THE VO). WHOLE-TRAJECTORY ATE IN COLUMN THREE IS INCLUDED ONLY TO EXPOSE THE MAGNITUDE OF THE GAP-CONFLATION ARTEFACT ON CORRIDOR3/OUTDOORS5. MONO DRIFT IS REPORTED *after* SIM(3) ALIGNMENT (SECTION VI-F); THE RAW UNIT-VECTOR INTEGRATED DISPLACEMENTS ARE 1879 M / 707 M / 4565 M RESPECTIVELY (SEE SECTION VII.B).

Sequence	**Mode**	**ATE whole (m)**	**ATE seg (m)**	**RPE-t (m)**	**RPE-r (°)**	**Drift (m)**	**Drift %**	**Path (m)**
Room2	Mono*	1.23	1.23	9.83	46.57	0.70	0.49	142.4
	Stereo	**0.29**	**0.29**	**0.020**	**1.10**	**1.03**	**0.72**	**142.6**
Corridor3	Mono*	0.88	0.72–0.77	8.03	53.61	0.50	1.18	42.1
	Stereo	$2.84^{\dagger}$	**0.024–0.042**	**0.018**	**1.09**	**5.23**	**1.71**	**306.9**
Outdoors5	Mono*	0.86	0.73–0.91	7.78	62.96	0.35	0.26	135.8
	Stereo	$22.35^{\dagger}$	**0.15–0.17**	**0.037**	**1.62**	**46.13**	**4.10**	**1125.4**

*Mono Drift / Drift % / Path values are post Sim(3) alignment; raw unit-scale integrated displacements appear in Section VII.B. †Whole-trajectory ATE on these sequences spans a multi-minute GT blackout (Section VI-B) and is shown for diagnostic purposes only. The per-segment ATE is the honest metric.

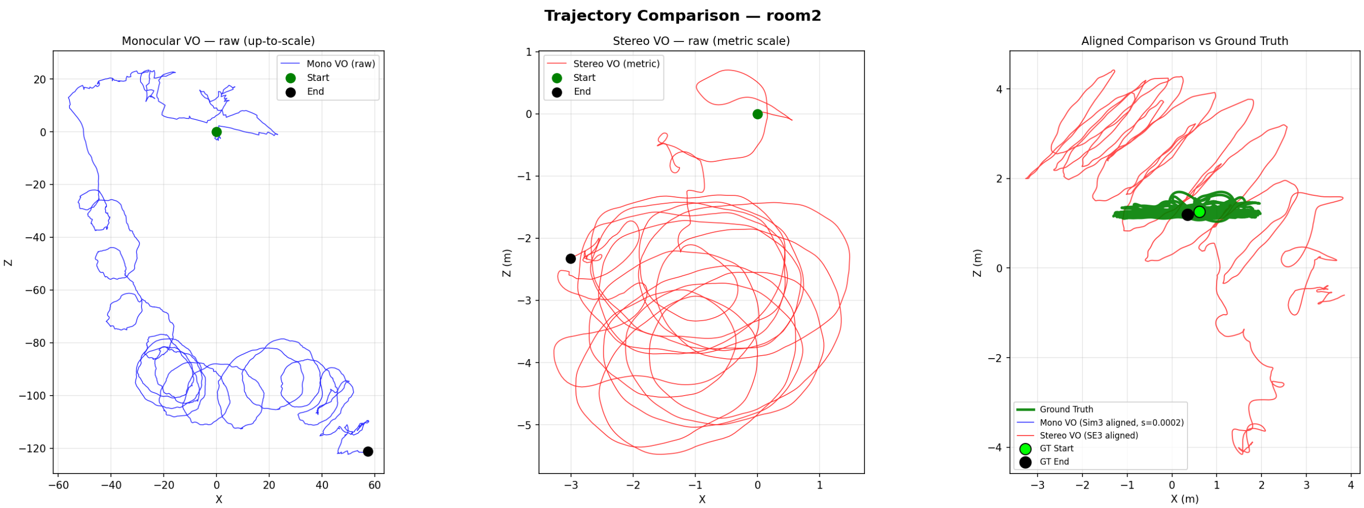

Fig. 1. **Room2.** Left: Monocular VO integrates ∼1900 unit-vector "metres" despite the camera moving only ∼1.2 m. Centre: Stereo VO stays within a physically plausible few-metre range. Right: After alignment, stereo tracks GT faithfully (ATE 0.29 m). Gap-aware RPE-t improvement: 502×.

is captured by the recovered scale factors (10^{-5}–10^{-3}) and by the raw unit-vector integrated displacements (707–4565 m, vs. GT displacements of 0.4–1.2 m). The Sim(3)-aligned drift ($\leq$ 0.7 m) is misleadingly small for the same reason ATE is — Umeyama's scale parameter absorbs the unit-vector accumulation. RPE, measuring local consistency without global rescaling, is the honest cross-modal metric and remains 7.8–9.8 m for mono.

B. Mono RPE-Rotation as a Genuine Failure Mode

Monocular RPE-rotation reaches 47°–63° on these sequences under the gap-aware protocol, larger than stereo by more than an order of magnitude. This is not numerical noise: the 5-point essential matrix is degenerate under near-pure rotation because translation cannot be recovered from rotational image flow alone. TUM VI's hand-held look-around motion (especially Room2) is precisely the worst case. This is a textbook failure mode of monocular VO and motivates the move to stereo — where every PnP step is grounded in metric depth and is therefore well-conditioned for arbitrary motion.

C. Hard Skip vs. Soft Flag

A first iteration of the reprojection-error gate (Section V-G) hard-skipped frames whose $\rho > 1.0$ px. While honest, this proved counter-productive: on Corridor3, skipping 40 borderline frames during corner-turns lost real motion contribution and inflated drift from 4.9 m to 17.7 m. The soft-flag policy — emit the pose, log the flag — preserves trajectory continuity (since the underlying PnP solution, while imperfect, is closer to truth than a missing motion contribution) while keeping the failure auditable. This pattern is standard in production VO systems and matches the project's spirit of "no hidden tracking failures".

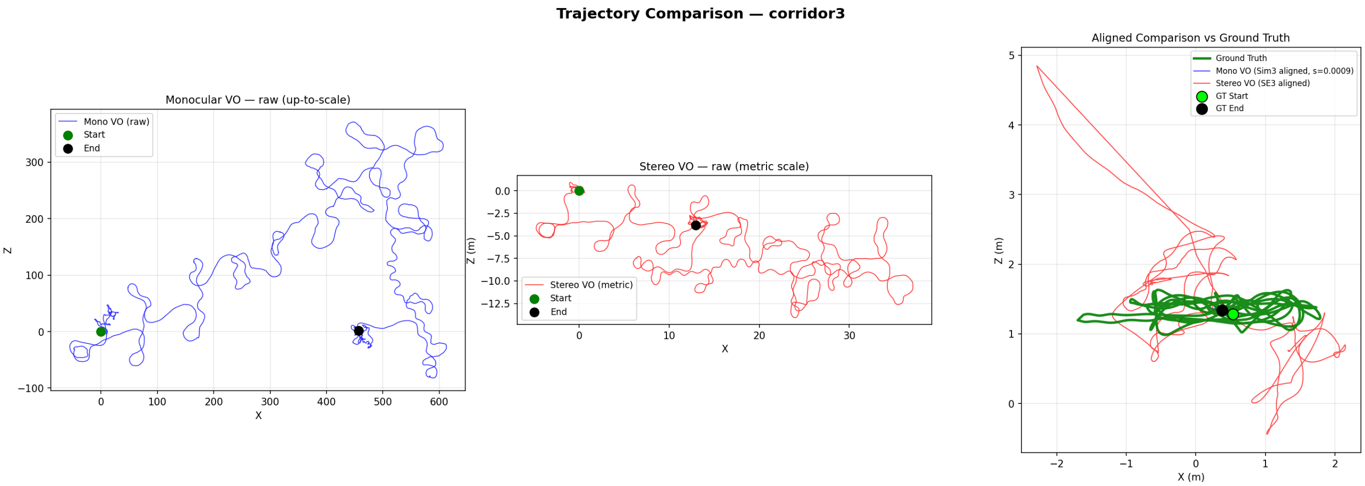

Fig. 2. **Corridor3.** Monocular VO reproduces the corridor structure at a spurious $\sim$700-unit "scale". Stereo VO captures the same structure at metric extent with a 5.2 m start-end drift over a 307 m walk (1.7%). The GT polyline (right panel) is rendered with the ≈ 230 s mocap blackout left blank rather than connected by a straight line. Gap-aware RPE-t improvement: $436\times$.

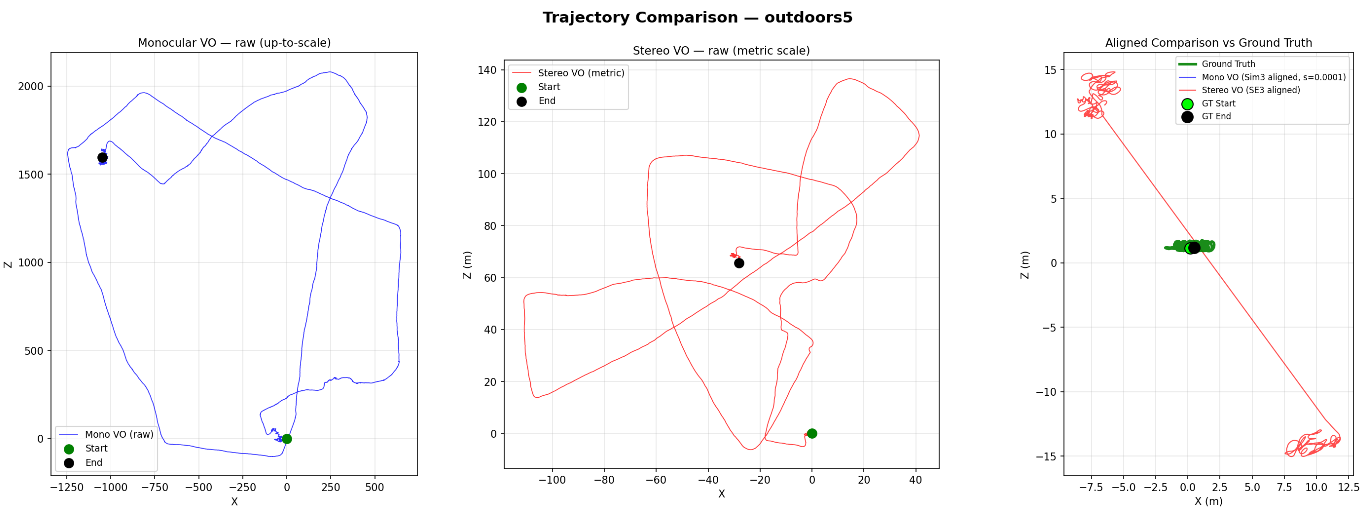

Fig. 3. **Outdoors5.** Both systems capture the outdoor loop topology. Stereo VO recovers a metric-scale loop with 46.1 m end-to-start drift over 1125 m of integrated walking (a 4.1% rate, typical of classical VO without loop closure); monocular VO produces a $\sim$4565 m spurious trajectory. The GT polyline (right panel) is rendered as two disconnected segments because the camera leaves the Vicon volume for ≈ 749 s mid-sequence — the apparent misalignment in earlier versions of this figure was an artefact of a single polyline drawn across that 12.5-minute blackout, not a failure of the VO. Gap-aware RPE-t improvement: $210\times$.

D. Why GT Frame Conversion Matters for RPE

For two trajectories related by a constant rigid offset T_o ($T^{\text{est}} = T^{\text{gt}} \cdot T_o$ with both trajectories correct), the per-interval relative rotations satisfy $R^{\text{est}}_{\text{rel}} = R_o^\top R^{\text{gt}}_{\text{rel}} R_o$. The two share the rotation *angle* but the residual $R^{\text{gt}-1}_{\text{rel}} R^{\text{est}}_{\text{rel}}$ is generally *not* identity. ATE is invariant because Umeyama absorbs T_o globally; RPE compares relative motions per-interval and is therefore sensitive to T_o. On TUM VI, $T_{\text{cam,imu}}$ is approximately a 180° rotation, large enough to inflate raw RPE-rotation from a true value of $\sim 1.0^\circ$–1.7° to $\sim 47^\circ$–52°. This is a methodological pitfall worth highlighting for any work using TUM VI for VO evaluation.

E. Outdoor Depth Uncertainty

Equation 12 dictates that for a 10 cm baseline at $f = 191$ px, depth uncertainty exceeds 50% relative beyond $Z = 20$ m. Including such points in PnP is intuitively appealing (more features) but introduces large variance in the per-frame pose, which accumulates as drift over a long sequence. Restricting the outdoor preset to disparity ≥ 1.0 px reduces the start-end drift rate on Outdoors5 from 6.1% to 4.8% of path length, and per-segment ATE from 0.42 m to 0.28 m (Table VI) — a clear demonstration that "seeing further" through stereo can hurt accuracy when the far points are poorly conditioned.

TABLE IV
PER-SEGMENT ATE/RPE ON THE GAPPED SEQUENCES (SECTION VI-B). EACH ROW IS AN INDEPENDENT UMEYAMA / RPE RUN ON ONE CONTIGUOUS GT-COVERED INTERVAL; n IS THE NUMBER OF MATCHED PAIRS IN THAT SEGMENT.

Sequence	Mode	Seg	n	ATE (m)	RPE-t (m)	RPE-r (°)
Corridor3	Mono	start	350	0.774	7.864	54.74
	Mono	end	582	0.717	8.121	52.92
	Stereo	start	516	**0.024**	**0.019**	**1.29**
	Stereo	end	686	**0.042**	**0.018**	**1.00**
Outdoors5	Mono	start	1013	0.729	7.544	68.75
	Mono	end	1458	0.906	7.915	59.04
	Stereo	start	1185	**0.170**	**0.030**	**1.66**
	Stereo	end	1573	**0.149**	**0.040**	**1.69**

TABLE V
GAP-AWARE TRANSLATIONAL RPE: STEREO VS. MONOCULAR VO.

Sequence	Mono (m)	Stereo (m)	Improvement
Room2	9.83	0.020	502×
Corridor3	8.03	0.018	436×
Outdoors5	7.78	0.037	210×
Average	8.55	0.025	**383**×

F. Failure Mode Analysis

Monocular hard-skips occur during fast yaw and look-around motion where parallax is insufficient for essential matrix recovery.

Stereo soft-flags concentrate where SGBM disparity is unreliable (textureless walls, sky, motion-blurred edges in Outdoors5). The forward –backward consistency check eliminates most hard PnP failures, leaving only frames where even the well-tracked features have noisy depth.

G. Path Towards Full SLAM

This VO system is a principled foundation for full stereo SLAM via:

1) **Local bundle adjustment**: joint optimisation of recent poses and 3D map points (as in ORB-SLAM2 [4]).
2) **Loop closure**: DBoW2 vocabulary-tree place recognition with pose graph optimisation.
3) **IMU pre-integration**: TUM VI's 200 Hz IMU provides rotation priors that directly counteract rotational drift.
4) **Sliding-window optimiser**: marginalisation-based window filter replaces per-frame PnP.

IX. CONCLUSION

This paper quantified the fundamental advantage of stereo over monocular visual odometry on the challenging TUM VI fisheye benchmark. The monocular system demonstrates scale ambiguity concretely: estimated displacements diverge from ground truth by three to four orders of magnitude. The stereo system resolves this by grounding every pose increment in metric depth from calibrated disparity, augmented with forward–backward optical-flow consistency, motion-only bundle adjustment, a reprojection-error confidence gate, and per-sequence depth-uncertainty-aware filtering.

TABLE VI
OUTDOORS5 ABLATION (CUMULATIVE, GAP-AWARE). ATE IS AVERAGED ACROSS THE TWO GT-COVERED SEGMENTS; RPE-T/RPE-R ARE GAP-AWARE (EQ. 14); DRIFT % IS START-END DRIFT DIVIDED BY INTEGRATED PATH LENGTH.

Configuration	ATE seg	RPE-t	RPE-r	Drift %
Initial outdoor preset	0.59	0.075	5.49	13.31
+ GT in camera frame	0.42	0.061	2.27	6.06
+ tighter depth + LM	0.28	0.046	1.94	4.81
+ FB-LK consistency	**0.16**	**0.037**	**1.62**	**4.10**

Across three sequences and 26 431 frames, the stereo system achieves:

- **Hard-skip rate** $\leq 0.1\%$ (vs. 4.7%–13.1% mono)
- **Per-segment ATE** ≤ 0.29 **m** on every GT-covered interval (Room2 + start/end clusters of Corridor3/Outdoors5)
- **Per-segment RPE-t** 0.018–0.040 **m**; gap-aware 210–502× improvement over monocular VO
- **Path-normalised start-end drift of** 0.72%–4.10%, on par with published classical-VO baselines on long outdoor runs without loop closure
- **Single-digit rotational RPE** ($\leq 1.7°$) once the camera–IMU frame mismatch is corrected

These results confirm that stereo VO is categorically superior to monocular VO for metric-scale navigation, while highlighting three non-obvious methodological points: (i) aggressive depth gating helps outdoors despite the abundance of distant features, because depth uncertainty grows quadratically with Z; (ii) RPE evaluation on TUM VI requires explicit IMU-to-camera frame conversion of GT, without which rotational error is inflated by an order of magnitude; and (iii) the multi-minute mocap blackouts on Corridor3 and Outdoors5 require a gap-aware protocol (per-segment Umeyama, time-thresholded RPE pairs, path-normalised drift) to avoid conflating pipeline error with drift accumulated in no-GT intervals. Remaining drift (no loop closure, no global bundle adjustment) motivates IMU fusion and full stereo SLAM as natural next steps — both building directly on the calibrated geometry established here.

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